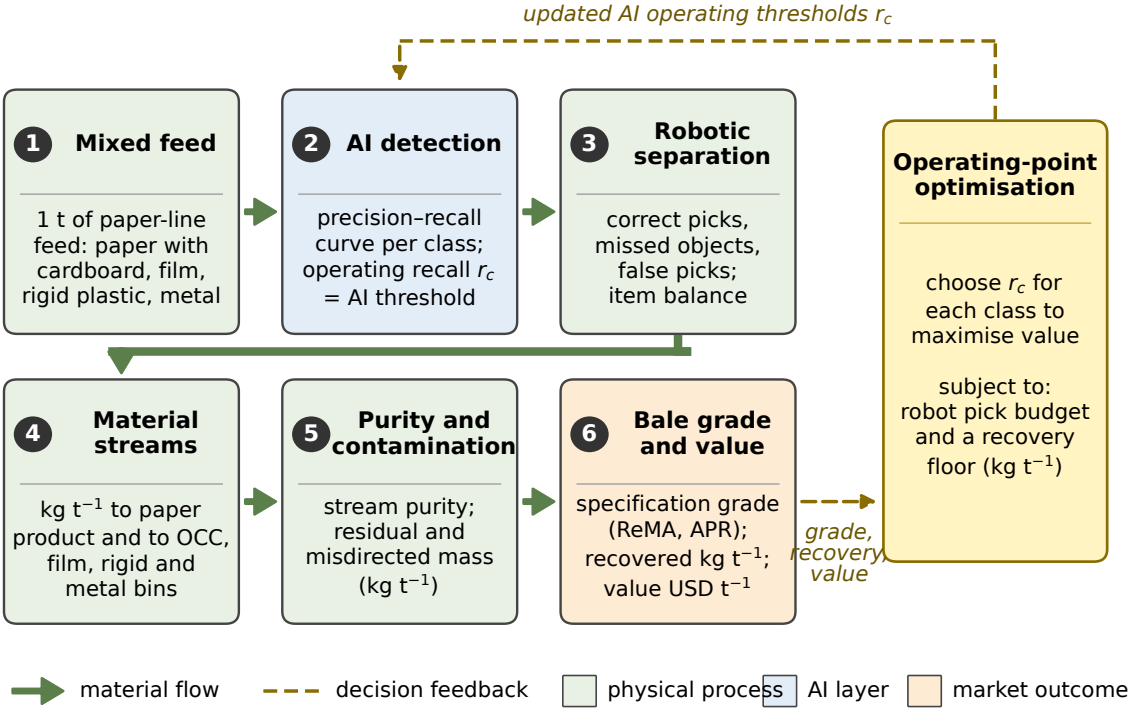
updated AI operating thresholds r_c
1 Mixed feed
1 t of paper-line feed: paper with cardboard, film, rigid plastic, metal
2 AI detection
precision–recall curve per class; operating recall r_c = AI threshold
3 Robotic separation
correct picks, missed objects, false picks; item balance
Operating-point optimisation
choose r_c for each class to maximise value
subject to: robot pick budget and a recovery floor (kg t^{-1})
4 Material streams
kg t^{-1} to paper product and to OCC, film, rigid and metal bins
5 Purity and contamination
stream purity; residual and misdirected mass (kg t^{-1})
6 Bale grade and value
specification grade (ReMA, APR); recovered kg t^{-1}; value USD t^{-1}
grade, recovery, value
material flow
decision feedback
physical process
AI layer
market outcome